

AI × KNOWLEDGE GRAPHS

Graph Mind

An **MCP server on Neo4j** that lets any AI tool query your knowledge graph — in plain English.

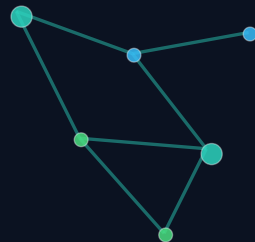
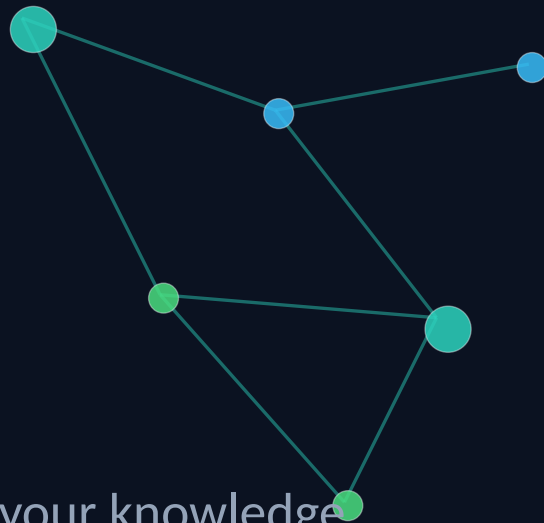
Natural language in. Graph intelligence out.

FastPath — no LLM

Read-only by design

Runs in VS Code

Hackathon Submission



The problem

Your richest data is relational — but it's trapped.



Locked in the graph

Org structure, skills, reporting lines, and collaboration live in graph databases only specialists can reach.



Cypher is a wall

Answering 'who is my manager's manager's manager?' means hand-writing graph queries. Business users can't.



AI can't plug in safely

Every AI tool needs bespoke, brittle glue — and giving an LLM raw DB access risks your data.

The solution

Graph Mind is an MCP server on Neo4j that turns natural language into safe graph answers.



Ask in plain English

No Cypher. FastPath turns common questions straight into graph queries — no LLM needed.



Plugs into your tools

Live in VS Code and any MCP AI tool, plus our own graph UI — all over HTTP.

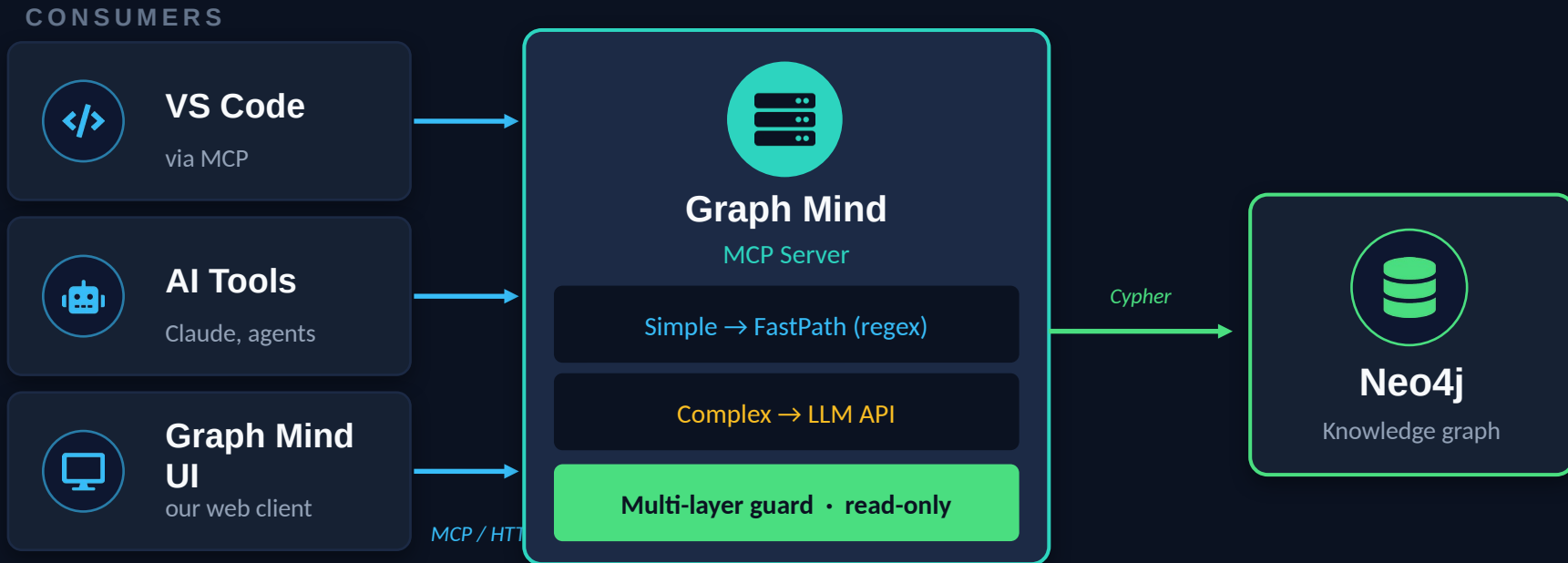


Safe & transparent

Read-only by design; every answer shows the exact Cypher it ran.

How it works

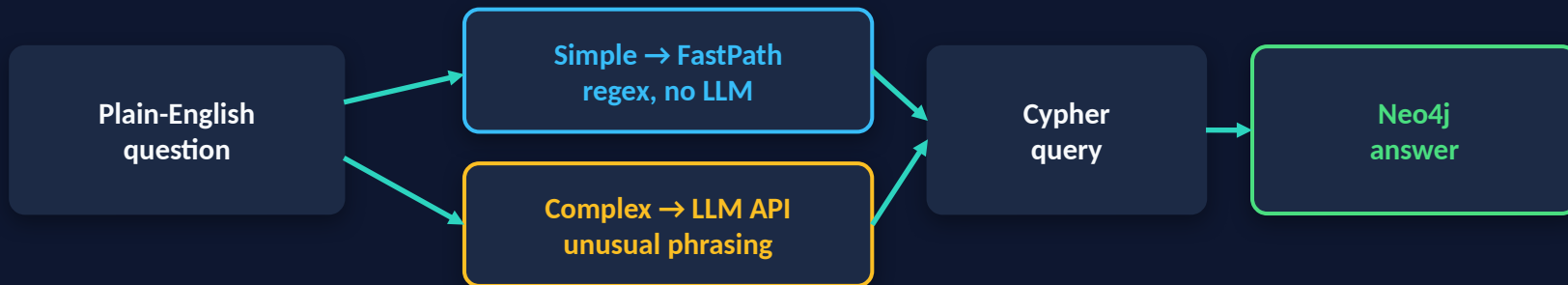
Everything talks HTTP. Graph Mind does the translation — and the guarding — in the middle.



Results return to the UI as an interactive graph + the exact Cypher.

FastPath: skip the LLM for simple queries

Simple questions become Cypher by regex — no model call. Complex ones use the LLM API.



No LLM, no cost

Simple queries never call an LLM — zero token cost.



Deterministic

Same question → same Cypher. Predictable, easy to test.



Read-only & safe

Patterns are bounded — no surprise or destructive queries.

Multi-layer security

Defense in depth — an AI that can read your graph and physically can't harm it.



LAYER 01

Keyword & injection scan

Incoming questions are scanned for dangerous keywords and prompt-injection patterns before anything runs.



LAYER 02

Read-only enforcement

Only MATCH / RETURN traversals execute. CREATE, MERGE, SET, DELETE and DROP are blocked at the server.



LAYER 03

Abuse throttling

Failed and blocked attempts are capped with retry limits and rate limiting — the guard can't be brute-forced.



LAYER 04

Transparent & auditable

Every answer ships with the exact Cypher it ran — so nothing is ever a black box.

What we built

A working end-to-end demo — in VS Code and in our own visual graph UI.

ASK ANYTHING



“Who is my manager’s manager’s manager?”



“Who shares a skill AND a project with Arjun?”



“Shortest path from me to the Payments lead?”



“Who can cover the Risk Architect role?”

Two surfaces, one server



VS Code — ask graph questions in your editor



Web UI — results render as an interactive graph



Cypher shown with every answer



FastPath + read-only under the hood



Live on a real employee graph in Neo4j

Why MCP changes the game

Build the bridge once — every AI tool can cross it.



One standard, not N×M glue

MCP is the protocol AI tools already speak. We implemented it once — VS Code worked for free.



Composable with agents

Graph Mind is a tool any agent can call mid-reasoning, blending graph facts with everything else.



Future-proof

As new AI clients adopt MCP, they get instant, safe access to your graph — no extra work for us.

Plug-in spotlight: Audience Builder

A notification composer builds send-lists straight from the graph — over UI or MCP.



For the iNOTIFY composer

Plug in via UI or MCP

Stop hand-curating recipient lists. The composer just asks — “everyone in APAC Payments reachable on Teams” — and Graph Mind returns the matching audience.

TARGET BY ATTRIBUTE

Region

Department

Role

Channel

Composer → Graph Mind (UI / MCP) → live audience set

ALSO UNLOCKS



Reach & de-dupe

Who a broadcast hits, duplicates collapsed across channels.



Escalation paths

Walk the chain of command to find who to notify next.



Always live

Audiences refresh as the graph changes — never stale.

Beyond employees

The same server works for any connected data. Employees were just the start.



Org intelligence

Reporting lines, span of control, team topology.



Skills discovery

Find who knows what, across the whole company.



Succession & impact

Who can cover a role; what breaks if someone leaves.



Collaboration graph

Who works with whom; hidden connectors.



Risk & fraud

Suspicious paths in transactions or access.



Any knowledge graph

Supply chains, infra, customers — plug it in.

The stack

Lean, standard pieces — wired together over HTTP.



Neo4j

Graph database storing connected data and running Cypher.



FastPath

Regex engine: NL → Cypher with no LLM call. LLM as fallback.



MCP Server

Multi-layer guard, over HTTP. Works in VS Code & any AI tool.



Web UI

Runs queries; renders results as a graph beside the Cypher.

THE VISION

Turn any knowledge graph into
something
an AI can reason over — safely, just by
asking.

WHAT'S NEXT

- Broaden FastPath pattern coverage
- Per-user auth & access control on the MCP layer
- Approved write actions — beyond read-only

Graph Mind — natural language in, graph intelligence out.

